Meridian Bank – Credit Card Churn Segmentation

Cards Analytics | Prepared for Rahul Menon, Head of Cards | Every number comes from the SQL file churn_analysis.sql

The short version. Meridian loses 19.46% of its cardholders. Customers who have **missed payments, hold only one product, are under financial stress, are unhappy with support, or keep contacting the bank** leave far more often than average. Customers with **no missed payments, no complaints and two or more products** almost never leave (4.7%). Card variant, time with the bank, credit limit, age and income make no real difference.

1. Baseline and definitions

Measure	Value
Total customers	100,000
Customers who closed their card (churned)	19,458
Portfolio churn rate (the baseline)	19.46%
High-churn line (1.5 times baseline or more)	29.19% or more
Low-churn line (0.6 times baseline or less)	11.67% or less
Smallest group we report (5% of customers)	5,000 customers
A "big difference" between two groups	20% or more, relative

- **Churn rate** of a group = customers who left ÷ customers in the group.
- **Churn share** of a group = customers who left from this group ÷ all 19,458 who left.
- Every group is compared with the 19.46% baseline, not with zero.

Things to know about the data

- The file has no "months inactive" field and no quarter-on-quarter change, so inactivity is measured with **monthly card transactions** and **monthly app logins**.
- "Contacts with the bank" means **customer service calls**. Complaints are tracked separately.
- Four scores that were already calculated by a model (churn probability, churn risk, engagement, customer value) were **left out**, because using them would be circular.
- There is no closure reason, so the results show **behaviour patterns, not proven causes**. Service calls, complaints and satisfaction may partly be a result of a customer deciding to leave, so treat them as early warnings.
- There is no profit data, so "is every departure worth preventing" can only be answered in general terms (see section 11).

2. How values were grouped into bands

Continuous numbers were cut into bands so groups are large enough to compare. Each rule is written so anyone can reproduce it.

Trait	Bands	Why
Age	Under 25, 25–34, 35–44, 45–54, 55–64, 65+	Standard life stages; 65+ is roughly retirement

Trait	Bands	Why
Time with bank	0–6, 7–12, 13–24, 25–48, 49–96, 97+ months	Fine bands early (first year, first renewal), wider later
Credit limit	Under 5k, 5–10k, 10–20k, 20–35k, 35k+	Every band holds at least 11% of customers
Utilisation	Under 20%, 20–40%, 40–60%, 60–80%, 80%+	Equal 20-point steps; 60%+ means a stretched card
Card transactions per month	0–10, 11–30, 31–60, 61–120, 121+	Measures inactivity; 0–10 is almost dormant
App logins per month	7 or fewer, 8–10, 11–13, 14–16, 17+	Second inactivity measure; median is 12
Service calls	0, 1, 2, 3–4, 5+	Whole numbers; rare high counts are grouped
Complaints	0, 1, 2, 3+	Same idea
Payment delays	0, 1, 2+	More than 2 is rare and behaves the same
Support satisfaction (1–5)	Below 2.5, 2.5–3.4, 3.5–4.4, 4.5+	Poor, middling, good, excellent
Financial stress score	Below 10, 10–19, 20–29, 30–39, 40–59, 60+	10-point steps; the very top is split off
Offers redeemed	0, 1, 2+	Whole numbers
Income	Under 30k, 30–60k, 60–100k, 100k+	Round cut-offs

Card variant, gender, marital status, education, employment and number of products use their own categories.

3. Churn by each trait on its own

Red rows are high-churn groups (29.19% or more); green rows are low-churn groups (11.67% or less). A star * means the group is under 5% of customers, so be careful with it.

Traits that separate leavers from stayers

Trait	Group	Customers	Churn rate %	Share of all churners %	Times average	Result
Payment delays	0 delays	70,059	12.5	44.9	0.64	average
	1 delay	20,785	27.0	28.9	1.39	average
	2+ delays	9,156	55.8	26.3	2.87	HIGH
Complaints	0	62,355	13.8	44.1	0.71	average
	1	23,439	22.2	26.7	1.14	average
	2	8,760	32.6	14.7	1.68	HIGH
	3+	5,446	51.7	14.5	2.66	HIGH
Service calls	0 service calls	39,947	13.8	28.3	0.71	average
	1	23,873	16.9	20.7	0.87	average
	2	14,482	20.9	15.6	1.08	average
	3-4	13,915	26.7	19.1	1.37	average
	5+	7,783	40.7	16.3	2.09	HIGH

Trait	Group	Customers	Churn rate %	Share of all churners %	Times average	Result
Support satisfaction (1–5)	<2.5	13,256	45.3	30.9	2.33	HIGH
	2.5-3.4	31,558	22.7	36.8	1.17	average
	3.5-4.4	43,121	12.4	27.6	0.64	average
	4.5+	12,065	7.7	4.8	0.40	LOW
Financial stress score	<10	8,217	6.4	2.7	0.33	LOW
	10-19	23,640	9.7	11.8	0.50	LOW
	20-29	29,226	14.1	21.2	0.73	average
	30-39	20,179	21.5	22.3	1.10	average
	40-59	14,967	36.7	28.2	1.89	HIGH
	60+ *	3,771	71.0	13.8	3.65	HIGH (small group)
Products held	1	45,035	25.9	59.9	1.33	average
	2	34,992	16.8	30.1	0.86	average
	3	15,040	10.8	8.3	0.55	LOW
	4 *	4,933	6.4	1.6	0.33	LOW (small group)
Credit utilisation	<20%	15,065	13.8	10.7	0.71	average
	20-40%	43,233	17.4	38.7	0.89	average
	40-60%	30,238	22.0	34.2	1.13	average
	60-80%	10,027	26.9	13.9	1.38	average
	80%+ *	1,437	35.0	2.6	1.80	HIGH (small group)
App logins per month	<=7	8,964	26.6	12.3	1.37	average
	8-10	25,794	22.5	29.8	1.16	average
	11-13	33,344	19.1	32.8	0.98	average
	14-16	21,753	16.2	18.2	0.83	average
	17+	10,145	13.3	6.9	0.68	average
Offers redeemed	0	35,000	23.3	41.9	1.20	average
	1	36,645	19.2	36.1	0.99	average
	2+	28,355	15.1	22.0	0.78	average

In simple words: the more payment delays, complaints, service calls and financial stress a customer has, and the lower their satisfaction and the fewer products they hold, the more likely they are to leave. Having 2+ payment delays is the strongest single signal (55.8% leave). Having 3 products or high satisfaction marks the safest customers.

Traits that do NOT separate leavers from stayers

Trait	Group	Customers	Churn rate %	Share of all churners %	Times average	Result
Card variant	Gold	18,061	19.2	17.9	0.99	average
	Platinum	7,123	18.9	6.9	0.97	average
	Silver	29,768	19.4	29.7	1.00	average
	Standard	45,048	19.6	45.5	1.01	average
Time with bank	0-6m	13,330	20.5	14.1	1.05	average
	7-12m	11,701	19.8	11.9	1.02	average
	13-24m	18,519	19.9	18.9	1.02	average
	25-48m	24,563	19.5	24.6	1.00	average
	49-96m	21,798	18.9	21.1	0.97	average
	97m+	10,089	18.0	9.3	0.93	average
Credit limit	<5k	11,635	20.7	12.4	1.06	average
	5-10k	24,222	19.3	24.0	0.99	average
	10-20k	32,942	19.2	32.5	0.99	average
	20-35k	19,571	19.4	19.5	1.00	average
	35k+	11,630	19.4	11.6	1.00	average

For these traits every group is close to the average. Lowest and highest group:

Trait	Lowest-churn group	Highest-churn group	Times average
Age	18.9% (65+)	20.1% (55-64)	0.97 to 1.03
Card transactions per month	17.2% (11-30)	21.8% (121+)	0.89 to 1.12
Income	19.3% (30-60k)	20.1% (<30k)	0.99 to 1.03
Gender	19.4% (Female)	19.7% (Other)	1.00 to 1.01
Marital status	19.2% (Widowed)	19.5% (Married)	0.99 to 1.00
Education	19.2% (Bachelor)	19.9% (High School)	0.99 to 1.02
Employment	19.1% (Self-Employed)	20.3% (Student)	0.98 to 1.04

In simple words: all of these sit within about 12% of the average, nowhere near the high-churn or low-churn lines. Notably, the **least active** card users churn at 17.7% while the **most active** churn at 21.8%, so "customers who barely use the card are the ones leaving" is not true in this data.

4. Shortlist of high-churn and low-churn groups

Only groups that clear the churn line **and** hold at least 5% of customers are listed.

Type	Group	Customers	Churn rate %	Share of all churners %	Times average
High churn	Payment delays: 2+ delays	9,156	55.8	26.3	2.87
High churn	Complaints: 3+	5,446	51.7	14.5	2.66
High churn	Support satisfaction (1–5): <2.5	13,256	45.3	30.9	2.33
High churn	Service calls: 5+	7,783	40.7	16.3	2.09
High churn	Financial stress score: 40-59	14,967	36.7	28.2	1.89
High churn	Complaints: 2	8,760	32.6	14.7	1.68
Low churn	Products held: 3	15,040	10.8	8.3	0.55
Low churn	Financial stress score: 10-19	23,640	9.7	11.8	0.50
Low churn	Support satisfaction (1–5): 4.5+	12,065	7.7	4.8	0.40
Low churn	Financial stress score: <10	8,217	6.4	2.7	0.33

No age, tenure, credit limit, card variant or demographic group reaches either line. Customers with only one product (25.9% churn) fall just below the high line, but they make up 59.9% of all customers who left, so they matter once combined with other traits.

5. The questions last year's review left open

Question	Group	Customers	Churn rate	Share of all churners	Answer
Do high-limit customers with no recent activity leave?	High limit (20k+) AND low card use (up to 30 card transactions a month)	890	15.1%	0.7%	No. Only 890 customers (0.9% of base, so low confidence) and they churn at 15.1%, below average.
	High limit (20k+) AND low app logins (up to 10)	10,768	23.4%	12.9%	Slightly higher (23.4%) but under the 1.5x high-churn line.
Do fee-paying premium cards leave faster?	Fee-paying variant (Silver/Gold/Platinum)	54,952	19.3%	54.5%	No. 19.3% vs 19.6% for Standard.
	No-fee variant (Standard)	45,048	19.6%	45.5%	
Do customers who contact the bank behave differently from silent ones?	Contacts: 3+ service calls	21,698	31.7%	35.4%	Yes. Customers who call 3+ times leave 2.3 times as often as silent ones.
	Silent: 0 service calls	39,947	13.8%	28.3%	Silent customers are the safer group, not the riskier one.
Does time with the bank change things?	Tenure up to 12m	25,031	20.2%	25.9%	Barely. 20.2% vs 18.0% is a 12% gap, below the 20% "big difference" bar.
	Tenure 97m+	10,089	18.0%	9.3%	
Does it change inside fee-paying cards?	Fee variant AND tenure up to 12m	13,627	20.2%	14.1%	Same pattern: 20.2% vs 18.0%.
	Fee variant AND tenure 97m+	5,555	18.0%	5.1%	

The review also asked whether every departure is worth preventing. Without profit data this cannot be measured; see section 11.

6. Combined groups (two or three traits together)

Single traits rarely describe a real customer, so these groups combine traits. Each one is defined by a rule anyone can re-run and holds at least 5% of customers. The groups overlap, so their shares must **not** be added together.

ID	Rule	Customers	% of base	Churn rate %	Share of all churners %	Times average
H1	Only 1 product AND satisfaction below 2.5	6,002	6.0	54.6	16.8	2.81
H2	Only 1 product AND financial stress 40 or more	8,440	8.4	54.2	23.5	2.79
H5	Only 1 product AND satisfaction below 3.5 AND financial stress 30+	9,136	9.1	53.0	24.9	2.72
H4	1+ payment delays AND 3+ service calls	6,533	6.5	50.5	17.0	2.60
H3	1+ payment delays AND only 1 product	13,427	13.4	45.2	31.2	2.32
H6	3+ service calls AND 2+ complaints	11,891	11.9	41.6	25.4	2.14
L5	Financial stress below 30 AND 3+ products	12,258	12.3	4.5	2.8	0.23
L1	No payment delays AND no complaints AND 2+ products	24,042	24.0	4.7	5.8	0.24
L4	No service calls, no delays, no complaints AND 2+ products	15,449	15.4	4.7	3.8	0.24
L3	Financial stress below 20 AND satisfaction 3.5+	19,332	19.3	5.3	5.3	0.27
L2	Financial stress below 20 AND 2+ products	17,523	17.5	5.5	5.0	0.28

Pairs of traits that were also tested point the same way. The sharpest high-churn pairs: only 1 product with satisfaction below 2.5 (54.6%), only 1 product with financial stress 40–59 (47.7%), and 2+ payment delays with **zero** complaints (47.5%, 5,660 customers). The last one shows payment trouble alone is enough, since these customers are not even complaining. The safest pairs: financial stress below 10 with zero complaints (3.5%) and 3 products with no payment delays (5.7%).

7. High-churn group vs low-churn group, side by side

High-churn group (H3): at least 1 payment delay AND only 1 product. 13,427 customers, 45.2% churn.
Low-churn group (L1): no payment delays, no complaints, 2 or more products. 24,042 customers, 4.7% churn.
The high group leaves about **9.6 times** as often. Highlighted rows differ by 20% or more.

Trait	High-churn group	Low-churn group	Difference	Big difference?
Average age	43.53	43.62	0.2%	No
Average months with bank	42.66	42.41	0.6%	No
Average credit limit	17,932	17,845	0.5%	No
Average income	80,272	80,477	0.3%	No
Average credit score	688.89	689.91	0.1%	No
Average utilisation	0.38	0.38	0.1%	No
Average monthly spend	6,130	6,085	0.7%	No
Card transactions per month	72.77	72.97	0.3%	No
App logins per month	12.06	11.99	0.6%	No
Website visits per month	5.99	5.99	0.1%	No
Service calls	1.50	0.56	169.3%	Yes
Complaints	0.60	0.00	Low group is zero	Yes
Payment delays	1.44	0.00	Low group is zero	Yes
Late-payment amount	66.28	0.00	Low group is zero	Yes
Support satisfaction (1–5)	3.17	3.99	25.9%	Yes
Financial stress score	43.32	21.74	99.2%	Yes
Products held	1.00	2.45	145.1%	Yes
Annual fee	75.48	74.78	0.9%	No
Interest rate	16.67	16.66	0.1%	No
Offers received	3.02	3.00	0.7%	No
Offers redeemed	1.06	1.05	0.7%	No
Share with a loan	34.8%	35.1%	1.0%	No
Share with a savings account	59.9%	59.6%	0.4%	No
Reward points	7,044	7,015	0.4%	No
Cash withdrawals	1.19	1.21	1.3%	No
International transactions	6.24	6.29	0.8%	No
Churn rate	45.2%	4.7%	870.1%	Yes

In simple words: the two groups look the same on who they are (age, income, credit limit, time with bank, spending, card use, app use, fees, offers; all within about 1.5%). They differ only on **financial distress** (stress score, payment delays, late amounts), **friction with the bank** (service calls, satisfaction) and **depth of relationship** (number of products). Complaints, payment delays and products differ partly because the group rules are built on them. A targeting rule based on age, tenure, limit or card tier could not tell these two groups apart.

8. Deep dive 1: a high-churn group (payment delay AND only 1 product)

The group is split step by step: first by **satisfaction**, then by **service calls**, then by **utilisation**. Starting point: 13,427 customers, 45.2% churn, 31.2% of all customers who left. Groups under 1,000 customers are marked low confidence.

Step	Group	Customers	Churned	Churn rate %	Share of all churners %	Confidence
Whole group	ALL: payment delays>=1 + 1 product	13,427	6,068	45.2	31.2	
Split 1	satisfaction <2.5	2,664	1,848	69.4	9.5	
Split 1	satisfaction 2.5-3.4	5,246	2,473	47.1	12.7	
Split 1	satisfaction 3.5+	5,517	1,747	31.7	9.0	
Split 2	satisfaction <2.5 → service calls 0-2	1,117	721	64.5	3.7	
Split 2	satisfaction <2.5 → service calls 3+	1,547	1,127	72.9	5.8	
Split 2	satisfaction 2.5-3.4 → service calls 0-2	4,339	1,999	46.1	10.3	
Split 2	satisfaction 2.5-3.4 → service calls 3+	907	474	52.3	2.4	Low (small group)
Split 2	satisfaction 3.5+ → service calls 0-2	5,075	1,588	31.3	8.2	
Split 2	satisfaction 3.5+ → service calls 3+	442	159	36.0	0.8	Low (small group)
Split 3	satisfaction <2.5 → service calls 0-2 → utilisation <40%	652	390	59.8	2.0	Low (small group)
Split 3	satisfaction <2.5 → service calls 0-2 → utilisation 40%+	465	331	71.2	1.7	Low (small group)
Split 3	satisfaction <2.5 → service calls 3+ → utilisation <40%	915	631	69.0	3.2	Low (small group)
Split 3	satisfaction <2.5 → service calls 3+ → utilisation 40%+	632	496	78.5	2.5	Low (small group)
Split 3	satisfaction 2.5-3.4 → service calls 0-2 → utilisation <40%	2,554	1,048	41.0	5.4	
Split 3	satisfaction 2.5-3.4 → service calls 0-2 → utilisation 40%+	1,785	951	53.3	4.9	
Split 3	satisfaction 2.5-3.4 → service calls 3+ → utilisation <40%	537	261	48.6	1.3	Low (small group)
Split 3	satisfaction 2.5-3.4 → service calls 3+ → utilisation 40%+	370	213	57.6	1.1	Low (small group)
Split 3	satisfaction 3.5+ → service calls 0-2 → utilisation <40%	2,961	759	25.6	3.9	
Split 3	satisfaction 3.5+ → service calls 0-2 → utilisation 40%+	2,114	829	39.2	4.3	
Split 3	satisfaction 3.5+ → service calls 3+ → utilisation <40%	248	77	31.0	0.4	Low (small group)

Step	Group	Customers	Churned	Churn rate %	Share of all churners %	Confidence
Split 3	satisfaction 3.5+ → service calls 3+ → utilisation 40%+	194	82	42.3	0.4	Low (small group)

What this shows: churn rises at every step. Low satisfaction lifts the group from 45.2% to **69.4%**. Adding 3+ service calls takes it to **72.9%**, and adding high utilisation takes it to **78.5%** (only 632 customers, so low confidence). No sub-group drops below the 19.46% average: even the calmest one (satisfied, few calls, utilisation under 40%, 2,961 customers) still churns at **25.6%**. A missed payment on a customer's only product is the main risk, and poor service makes it worse.

9. Deep dive 2: a low-churn group (no delays, no complaints, 2+ products)

The group is split step by step: first by **products held**, then by **financial stress**, then by **satisfaction**. Starting point: 24,042 customers, 4.7% churn, only 5.8% of all customers who left.

Step	Group	Customers	Churned	Churn rate %	Share of all churners %	Confidence
Whole group	ALL: no delays + no complaints + 2+ products	24,042	1,120	4.7	5.8	
Split 1	2 products	15,359	894	5.8	4.6	
Split 1	3-4 products	8,683	226	2.6	1.2	
Split 2	2 products → financial stress <20	6,865	261	3.8	1.3	
Split 2	2 products → financial stress 20-29	5,388	334	6.2	1.7	
Split 2	2 products → financial stress 30+	3,106	299	9.6	1.5	
Split 2	3-4 products → financial stress <20	3,830	61	1.6	0.3	
Split 2	3-4 products → financial stress 20-29	3,062	76	2.5	0.4	
Split 2	3-4 products → financial stress 30+	1,791	89	5.0	0.5	
Split 3	2 products → financial stress <20 → satisfaction <3.5	1,275	64	5.0	0.3	
Split 3	2 products → financial stress <20 → satisfaction 3.5+	5,590	197	3.5	1.0	
Split 3	2 products → financial stress 20-29 → satisfaction <3.5	942	72	7.6	0.4	Low (small group)
Split 3	2 products → financial stress 20-29 → satisfaction 3.5+	4,446	262	5.9	1.3	
Split 3	2 products → financial stress 30+ → satisfaction <3.5	520	65	12.5	0.3	Low (small group)
Split 3	2 products → financial stress 30+ → satisfaction 3.5+	2,586	234	9.0	1.2	

Step	Group	Customers	Churned	Churn rate %	Share of all churners %	Confidence
Split 3	3-4 products → financial stress <20 → satisfaction <3.5	696	18	2.6	0.1	Low (small group)
Split 3	3-4 products → financial stress <20 → satisfaction 3.5+	3,134	43	1.4	0.2	
Split 3	3-4 products → financial stress 20-29 → satisfaction <3.5	549	25	4.6	0.1	Low (small group)
Split 3	3-4 products → financial stress 20-29 → satisfaction 3.5+	2,513	51	2.0	0.3	
Split 3	3-4 products → financial stress 30+ → satisfaction <3.5	339	22	6.5	0.1	Low (small group)
Split 3	3-4 products → financial stress 30+ → satisfaction 3.5+	1,452	67	4.6	0.3	

What this shows: this group is safe because several protective factors stack up. Each extra layer (more products, lower stress, higher satisfaction) lowers churn further, down to just **1.4%** for customers with 3–4 products, stress below 20 and satisfaction 3.5+ (3,134 customers). Even the weakest sub-group (2 products, stress 30+, satisfaction below 3.5: 12.5%, 520 customers, low confidence) is still below the average. These customers do not need retention offers.

10. Findings

1. Missed payments are the strongest warning sign.

Customers with 2+ payment delays churn at 55.8% (2.87 times the average) and are 9.2% of customers but 26.3% of all churn. With only one product, any delay gives 45.2% churn (13.4% of customers, 31.2% of all churn).

What it means for retention: Reach out at the first missed payment with payment help or a restructured plan, before any fee waiver or bonus offer.

2. Single-product customers under financial stress are the biggest leak.

One product plus a financial stress score of 40+ means 54.2% churn, and this group holds 23.5% of all churn from 8.4% of customers. Customers with 3 products churn at only 10.8%.

What it means for retention: Getting a second product (savings account, loan) is the strongest lasting retention lever. Offer it before stress appears.

3. Customers who contact the bank leave more; silent customers stay.

5+ service calls means 40.7% churn and satisfaction below 2.5 means 45.3%, against 13.8% for customers who never call.

What it means for retention: Treat a service call, complaint or low satisfaction score as a trigger for a personal save attempt. Silence is not a warning sign.

4. The usual targeting traits do not predict churn.

Fee cards churn at 19.3% vs 19.6% for Standard. Time with the bank ranges only 18.0%–20.5%. Credit limit and age are flat, and high-limit, low-usage customers churn at 15.1% (890 customers).

What it means for retention: Fee waivers and blanket "inactive customer" lists, which drove last year's campaign, were aimed at the wrong people.

5. **Churn is concentrated and clearly separable, about a 10 times gap.**

   Payment delay plus one product churns at 45.2%, against 4.7% for customers with no delays, no complaints and 2+ products (24% of customers, 5.8% of churn), yet the two groups look identical on age, tenure, limit, income and spend.

   What it means for retention: Leave the clean, multi-product 24% out of campaigns and focus budget on the roughly 13% of customers who carry 31% of churn.

11. Limits of this analysis

- **Cause is not proven.** The data has no exit reason. Service calls, complaints and satisfaction may follow a decision to leave rather than cause it. Use them as early warnings.
- **Not every departure is worth preventing.** Customers with a financial stress score of 60+ (71.0% churn, 3.8% of customers, low confidence) and repeated delays may be credit risks whose exit is partly healthy. No profit data was supplied, so before designing offers, add expected profit (interest income minus expected losses) to the high-churn groups. For the same reason, tenure cannot be linked to the cost of a departure.
- **Inactivity was measured indirectly** through card transactions and app logins. Months since last use and quarter-on-quarter changes were not available and could still hold a signal.
- **The data is a single snapshot,** so behaviour in the months before leaving cannot be seen.
- **Groups overlap.** The combined groups are different cuts of the same distress pattern; do not add their shares.